

Bank Loan Report - Project Documentation

1. Problem Statement

In order to monitor and assess our bank's lending activities and performance, we need to create a comprehensive Bank Loan Report. This report aims to provide insights into key loan-related metrics and their changes over time. The report will help us make data-driven decisions, track our loan portfolio's health, and identify trends that can inform our lending strategies.

Dashboard 1: Summary

Total Loan Applications: Calculate the total number of loan applications received during a specified period. Monitor the Month-to-Date (MTD) Loan Applications and track changes Month-over-Month (MoM).

Total Funded Amount: Understand the total amount of funds disbursed as loans. Monitor the MTD Total Funded Amount and analyse the Month-over-Month (MoM) changes.

Total Amount Received: Track the total amount received from borrowers to assess cash flow and loan repayment. Analyse the Month-to-Date (MTD) Total Amount Received and observe the Month-over-Month (MoM) changes.

Average Interest Rate: Calculate the average interest rate across all loans, MTD, and monitor the Month-over-Month (MoM) variations to provide insights into the lending portfolio's overall cost.

Average Debt-to-Income Ratio (DTI): Evaluate the average DTI for borrowers to gauge their financial health. Compute the average DTI for all loans, MTD, and track Month-over-Month (MoM) fluctuations.

Good Loan v Bad Loan KPIs

To evaluate the performance of our lending activities and assess portfolio quality, we distinguish between 'Good Loans' and 'Bad Loans'.

- **Good Loan KPIs:** Includes loans with a status of 'Fully Paid' and 'Current'. Metrics include Application Percentage, Total Applications, Funded Amount, and Total Received Amount.
- **Bad Loan KPIs:** Includes loans with a status of 'Charged Off'. Metrics include Application Percentage, Total Applications, Funded Amount, and Total Received Amount.

Loan Status Grid View

A grid view report categorized by 'Loan Status' provides a comprehensive overview of operations. It tracks Total Loan Applications, Funded Amount, Amount Received, MTD Funded Amount, MTD Amount Received, Average Interest Rate, and Average DTI by status.

2. Domain Knowledge

Bank loans are a crucial financial tool that enables individuals and businesses to achieve their goals. Understanding the terms, costs, and responsibilities associated with loans is essential for informed financial decisions.

Data Collection Channels

- **Loan Applications:** Detailed applications submitted electronically or in paper form.
- **Credit Reports:** Accessed from credit bureaus to assess creditworthiness, history, and payment behaviour.
- **Internal Records:** Maintained in the bank's database, covering disbursements, repayments, and status changes.
- **Online Portals & Third-party Sources:** Portals for application and payment, alongside external income verification services.

Process of Granting a Loan

1. **Loan Application & Review:** Customer submits an application, and the institution reviews necessary documentation.
2. **Verification & Credit Check:** Identity, credit history, income, and employment are verified. Debt-to-Income Ratio (DTI) is calculated.
3. **Risk Assessment & Approval:** Lenders evaluate the risk. If approved, the loan amount, rate, and terms are established.

4. **Agreement & Disbursement:** The borrower signs the agreement, and funds are disbursed.
5. **Repayment & Monitoring:** The borrower makes regular payments, and the lender continuously monitors the loan's health.

Reasons for Analysing Bank Loan Data

- **Risk Assessment & Credit Risk Management:** Evaluate creditworthiness, predict default probabilities, and manage overall credit risk.
- **Decision-making & Portfolio Management:** Support lending decisions and optimize portfolio health and pricing.
- **Fraud Detection & Regulatory Compliance:** Detect unusual patterns and comply with regulations like HMDA and KYC.
- **Customer Insights & Retention:** Understand behaviour, tailor products, and identify refinancing opportunities.
- **Profitability Analysis & Market Research:** Assess portfolio profitability and understand market trends.